Liquidity, Activism Disclosure, and Takeover Premia

A Theory of Exit, Voice, and Corporate Control

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Abstract

A blockholder who crosses five percent of a voting class must reveal herself on Schedule 13D, and since February 2024 that filing is due in five business days rather than ten calendar days. I study how this legal disclosure clock — a stake threshold plus a filing window — shapes what the market knows while a stake is being built, and what that means for corporate control. In the model, the blockholder first trades without a public flag; once her stake crosses the threshold and the filing arrives, the market sees her stake and intent. This partition of histories into flagged and pooled cells gives the threshold and the window different economic roles. The pooled cell carries all of the liquidity sensitivity of the engagement-related premium, while the flagged endpoint is insulated from noise-trading intensity. At fixed trading policies, a tighter threshold reduces that sensitivity unconditionally; a shorter window does so only when the change in who remains hidden is not too large. The clock binds in practice: in a near-census of initial Schedule 13D filings from mid-2023 and late 2024, the median trigger-to-filing delay fell from six business days to five, and the share filed within five business days rose from 38.9 to 67.1 percent — a 28.2-percentage-point difference, a descriptive comparison with no control group. The model gives no unconditional sign for the 2024 US acceleration, and the paper estimates no causal effect on returns, activism, or takeovers.

1 Introduction

A blockholder who learns that a company is worth more than the market price can sell, wait, or try to change the company. If she builds a large enough position while she waits, the law makes her disclose it. In the United States, Section 13(d) of the Williams Act requires a holder who crosses 5 percent of a voting class to file a Schedule 13D. Until February 2024 the initial deadline was ten calendar days; it is now five business days (U.S. Securities and Exchange Commission, 2023). That rule gives me two policy variables. The threshold determines which positions become public. The window determines how much trading can take place before the filing. A lower threshold and a shorter window can therefore change different parts of the market's information.

The central point is that disclosure divides the histories seen by the bidder and by market makers. After a filing, the market sees the blockholder's stake and her intention to engage. Before a filing, it sees order flow and must infer whether an informed holder is trading. I call these the flagged and pooled cells. The distinction matters because a bidder responds to both the expected value of the firm and the chance that engagement has raised it. Liquidity changes the inference in the pooled cell, but it need not change what the market learns from a completed filing.

Several literatures contain pieces of this argument. Liquidity can finance monitoring (Maug, 1998), have no effect under free entry (Kahn and Winton, 1998), or weaken the information in an exit threat (Admati and Pfleiderer, 2009). Trading models study pooled and revealed orders without a legal disclosure rule (Kyle and Vila, 1991). With a fixed disclosure horizon, shortening the horizon can be equivalent to reducing noise-trading volatility (Back et al., 2018); other work places a multi-trader game inside a pre-disclosure window (Cetemen et al., 2026). None of these papers combines the two legal margins with liquidity and an endogenous control outcome. That combination is the contribution here.

The model has three main implications. First, a cutoff equilibrium over complete trading plans exists under conditions on the plan menu, prices, and the best-response map. The exact construction and its proof are in the online appendix. Two of the conditions fail at the implemented calibration, so the result does not establish existence there. Twenty-three of twenty-seven numerical sweep nodes converge, and the other four remain unresolved. I make no nonexistence claim.

Second, at fixed trading policies, the expected engagement-related premium has a simple decomposition: only the pooled cell contributes to its liquidity sensitivity. A tighter threshold moves histories into the flagged cell and therefore reduces that sensitivity. A shorter window has a weaker result. It reduces the weight on the pooled cell, but it also changes which histories remain there. The window reduces sensitivity precisely when the weight effect dominates that composition effect. This is the fixed-policy result stated in Theorem 1.

Third, allowing the blockholder's policy to respond to liquidity adds a feedback term. The fixed-policy sign survives for the threshold margin on a region where the best-response map contracts and the direct effect dominates the feedback bound. The region is a condition of the result, not an object I claim to have found. Eighteen of eighty grid nodes satisfy the pointwise inequalities, with a largest contiguous block of nine. Proposition 2 states this implication.

The calibration gives a useful but limited guide to the 5 February 2024 filing-window change. Holding policies fixed and shortening the window from ten to five days reduces the computed sensitivity to between 18 and 77 percent of its long-window value at every checked node. The composition effect accounts for most of the reduction. At the same calibration, however, the pooled posterior has 23 to 767 distinct values at each of 180 non-degenerate nodes, rather than the three-point support required by the threshold comparison. The directional window result is therefore numerical evidence, not a theorem about the implemented two-round model.

The filing data show that the clock binds in practice. Comparing the near-census of initial Schedule 13D filings from 2023Q2–Q3, ahead of the rule's adoption, with 2024Q3–Q4, after the new deadline took effect, the median trigger-to-filing delay fell from six business days to five, and the share filed within five business days rose from 38.9 to 67.1 percent — a difference of 28.2 percentage points, with a 95 percent cluster-bootstrap interval of 21.8 to 34.8. The comparison is descriptive and has no control group; it establishes that the statutory clock moved realised filing timing, and it identifies no effect on liquidity, returns, or control outcomes.

Section 2 describes the rule and the literature. Section 3 lays out the two-round game in words and notation. Section 4 records the equilibrium construction briefly; the technical argument is in the appendix. Section 5 gives the two comparative-statics results and the calibration record. Section 6 presents the filing-delay evidence and the audit behind it. Section 7 closes. The proofs sit in the online appendix.

2 Institutional setting and related literature

2.1 The disclosure rule and its two margins

Section 13(d) of the Williams Act of 1968 requires any person or group acquiring beneficial ownership of more than 5 percent of a class of registered voting equity to file a Schedule 13D disclosing the stake and its purpose. From 1968 until 2024 the initial deadline was ten calendar days. The SEC proposed shortening it in February 2022, adopted the amendments in October 2023, and made them effective February 5, 2024: the initial 13D deadline became five business days, amendments on material change became two business days, and the parallel changes for Schedule 13G took effect September 30, 2024 (U.S. Securities and Exchange Commission, 2022; U.S. Securities and Exchange Commission, 2023). Item 4 of the form asks the filer to state its purpose; control-intent filers use the 13D, while the shorter 13G is open to three eligible classes (exempt investors, qualified institutional investors, and passive holders below 20 percent who certify no control intent). Which of the two a given block lands in is a choice the rule frames rather than an accident of the holder, and eligibility for the short form is itself a type variable rather than a random one.

Both margins of the rule vary across jurisdictions, and the variation is not subtle. The United Kingdom sets the initial threshold at 3 percent with notification at every additional percentage point, within two trading days. The EU Transparency Directive sets a ladder at 5 percent and above with four trading days, and several member states start at 3 percent. The threshold margin is therefore a cross-section object and the window margin a time-series one. Other jurisdictions have moved their windows too, the EU cutting its Transparency Directive deadline to four trading days in 2013 among them; what recommends the February 2024 acceleration as an anchor is that it is a dated change to one margin, in one jurisdiction, with a clean before and after. That is why it anchors this paper and why it has begun to anchor others (Polk et al., 2024; Trivedi, 2026).

Two institutional facts discipline the model. First, the window is a *legal deadline attached to a partition*, not a random horizon: the clock starts when the stake crosses the threshold, the filing lands at most T days later, and between the crossing and the filing the market prices the stock without the flag. The model takes the deadline itself as the filing date, so filing early is outside what it claims. Second, the pooled cell is pooled *for the price-setting market*. Zeng (2026) documents that a target's own managers learn of the accumulation through stock-surveillance vendors before the filing, and that the price run-up begins on the trigger date. The cell is leaky at its edges, which is a measurement problem for the empirics, not a failure of the partition for the market makers who set prices. On the delay distribution itself, Bebchuk et al. (2013) show filers bunching at the old deadline: 45.3 percent of filings by engaging investors in their sample land in the eight-to-ten-*business*-day bucket and roughly a fifth on the tenth business day itself, their delays being counted in business days against a deadline the rule states in calendar days. So the window binds for a large share of filers, which is what makes the February 2024 acceleration a treatment rather than a formality.

2.2 Exit, voice, and liquidity

The classic trade-off originates with Hirschman (1970). Coffee (1991) and Bhide (1993) argue that liquidity weakens governance by letting blockholders sell instead of fight; Maug (1998) reverses the logic: liquid markets let a monitor recoup the costs of engagement through informed trading, and with an endogenous stake liquidity raises equilibrium monitoring. Kahn and Winton (1998) show the sign of the trading motive depends on whether engagement deepens the blockholder's information advantage; Faure-Grimaud and Gromb (2004) make price informativeness and insider incentives complements; Admati and Pfleiderer (2009) make exit itself a disciplining threat. On the empirical side, Edmans, Fang, et al. (2013) find more liquid targets shift from voice to exit, and Norli et al. (2015) find liquidity predicts the incidence of costly voice.

Two limitations of this strand shape the present paper. The four classics use four different liquidity parameters (camouflage, noise volume, informativeness, exit noise) with three different signs, so "liquidity and governance" is not one question. And none of the four has a bidder, an offer, or a premium: takeovers appear in Maug (1998) only as a relabelled intervention technology, Kahn and Winton (1998) and Faure-Grimaud and Gromb (2004) set control transfers aside in terms, and Admati and Pfleiderer (2009) simply has no bidder. The object this paper studies, the sensitivity of an expected takeover premium to liquidity and what the disclosure rule does to that sensitivity, is not stated in any of them.

2.3 Trading, disclosure, and the market for control

The trading model builds on discrete order flow in the Kyle tradition (Kyle, 1985; Glosten and Milgrom, 1985), in the tractable form of Edmans, Goldstein, et al. (2015), and on the feedback between prices and real decisions (Edmans, Goldstein, et al., 2012; Bond et al., 2012; Goldstein, 2023). Grossman and Hart (1980) make the takeover premium the welfare object for dispersed minority shareholders, and Burkart, Gromb, et al. (1998) show why higher premia protect them.

Against the closest theory papers the position is specific. Kyle and Vila (1991) have an endogenous pooled/revealed split in mixing equilibria but disclaim disclosure requirements on their first page, and their traders disclose nothing at a legal deadline; the defensible contrast is that their split is endogenous, un-keyed, and available only where the trader happens to mix, while the rule-keyed partition here is imposed by a stake trigger and a deadline in every parameter configuration.

Collin-Dufresne and Fos (2015) show measured price impact falls when informed blockholders trade against liquidity, and Collin-Dufresne and Fos (2016) fix a trading horizon, but the presence of the informed trader is common knowledge in their model; they name the inference problem and set it aside. Back et al. (2018) embed an engaged blockholder in a continuous-time Kyle model with a fixed horizon at which the stake is disclosed; for them a shorter horizon is a re-parameterisation of noise-trading volatility ("reducing the trading horizon T is isomorphic to reducing noise trading volatility"), there is no acquirer and no premium, and endogenizing the horizon is their own declared out-of-scope extension. Cetemen et al. (2026) study multiple blockholders timing trades and coordinating through market signals inside a pre-disclosure window they leave unmodelled (their own reading of their model), with liquidity the comparative static and no premium object. Corum (2025) has the closest reduced form of the rule, two scalars measuring how much of the stake is tradable before disclosure bites, but collapses threshold and window into them and has no second player. Corum and Levit (2019) put a takeover probability and a disclosure threshold in one model, but the threshold does three jobs at once (liquidity break, 13D trigger, pill trigger), no comparative static in it is stated, and the flagged state never occurs on path. Ordóñez-Calafí and Bernhardt (2022) study the optimal threshold level (crossing reveals the position, so in equilibrium the engaging blockholder's stake is the smaller of the threshold and its unconstrained optimum, the threshold acting as an upper bound on the position and hence on trading profits), and they assert, without a proposition, the interaction between liquidity and the threshold on managerial discipline; the object here is a control outcome and the interaction is proved. Burkart and Lee (2022) have the premium and explicitly hand over the missing piece: they do not endogenize the toehold in anonymous pre-disclosure markets and leave the interaction of pre- and post-disclosure decisions to future work.

2.4 Structural and empirical work on activism

A mature structural literature estimates activism along complementary margins: Gantchev (2013) the sequential costs of campaigns; Johnson and Swem (2021) a dynamic reputation model in which buying and filing are one move; Albuquerque et al. (2022) the binary 13D-versus-13G choice jointly with announcement returns, with liquidity a control variable and no control outcome; Celentano and Levine (2025) an equilibrium model of activism, takeovers, and managerial discipline in which the 5 percent threshold enters once, as a calibration constant, and the window never appears. None of these models the trading-and-disclosure environment itself. Brav et al. (2008) and Greenwood and Schor (2009) document that the targets of engagement campaigns are unusually likely to be acquired, and that fact makes bidder entry the natural control outcome; Becht et al. (2017) show the returns internationally; Zeng (2026) measures what leaks inside the window.

On the February 2024 acceleration itself, two papers precede this one and neither occupies the cell. Polk et al. (2024) use pre-rule data only, a cross-section of self-selected delays offered as a projection with no control group, no post-rule data, and no standard errors anywhere in the paper. Their treatment variable counts calendar days against a rule they themselves state in business days. Trivedi (2026) runs a difference-in-differences on 333 filings with 13G controls and finds a first stage on compliance timing (the share filed within five business days rises by 34.8 percentage points, standard error 13.0) and nulls on spread and Amihud changes that carry no minimum detectable effect; there is no control outcome in the paper, its sampling frame is never stated, and it is not peer reviewed. The cell this paper occupies (a rule margin, a liquidity split, and a control outcome) remains open. Section 6 reports the descriptive filing-delay evidence.

3 The model

3.1 Object and partition

The disclosure rule divides the market's information into two cases. A stake threshold τ and a filing window T determine whether the filing has arrived by the control decision. In the *flagged* cell, the market sees the stake and the intention to engage. In the *pooled* cell, it sees only order flow and must infer whether an informed blockholder is trading. The cells are exhaustive. The outcome of interest is the expected engagement-related premium $\Delta^{\text{act}}(\kappa, \tau, T)$. I also track the run-up before the filing and the jump when the filing arrives. A lower threshold is a tighter threshold margin; a shorter window is a tighter window margin.

The model uses the discrete order-flow structure of Kyle (1985) and Edmans, Goldstein, et al. (2015), competitive prices as in Glosten and Milgrom (1985), and the takeover premium as the dispersed shareholder's control-outcome measure in the spirit of Grossman and Hart (1980). The contribution is the legal partition between those familiar pieces.

3.2 Timing: the two rounds

Trading runs over business days $d = 0, \dots, H$. Nature draws the firm's standalone value and the blockholder's private signal. The blockholder then chooses a complete contingent plan from a finite ordered menu. The plan fixes the stake path and whether it is meant to engage the firm.

In the first round, market makers see the plan's order flow mixed with ternary noise. The blockholder does not revise the path in response to realised prices or order flow. The filing arrives exactly T business days after the first crossing of τ, provided that date is before the horizon. That filing ends the pooled round. In the second round the blockholder trades the residual position, the market updates its price, and the bidder decides whether to enter. If no filing arrives, the bidder acts on the pooled history. The technical appendix spells out the off-path beliefs and the regularity conditions needed for this timing.

3.3 Primitives

The firm's standalone value is $v \sim N(\mu_v, \sigma_v^2)$. The blockholder observes $s = v + \varepsilon$, where $\varepsilon \sim N(0, \sigma_\varepsilon^2)$ is independent of v. Her posterior mean is $\mathbb{E}[v \mid s] = \mu_v + \beta(s - \mu_v)$, with $\beta = \sigma_v^2/(\sigma_v^2 + \sigma_\varepsilon^2) \in (0, 1)$. A potential bidder has an independent synergy shock $\xi \sim N(0, \sigma_\xi^2)$, mean synergy $\bar{S}$, and entry cost $K > 0$. Engagement changes the takeover premium from m_0 to m_1, where $\Delta_m = m_1 - m_0 > 0$, and creates non-takeover value $\Delta_V \geq 0$.

Noise trading takes the values $-\bar{z}$, 0, and $+\bar{z}$. The zero-noise probability is $1 - \kappa$ and each nonzero value has probability $\kappa/2$, so $\kappa \in [0, 1]$ measures noise-trading intensity. The disclosure policy is (τ, T), with $T \in \{1, \dots, H\}$; b_0 and $\bar{b}$ are the initial and maximum stakes.

The blockholder chooses a plan from a finite menu $\mathcal{J}$, ordered from least to most aggressive. Plan j has an engagement flag a_j, equal to one for Voice and zero for Exit or Hold. Its pooled stake at date d is $B_j(s, d)$, with $B_j(s, -1) = b_0$, and its terminal target is $b_j^*(s) = B_j(s, H)$. The first crossing of the threshold is $c_j(s;\tau) = \inf\{d : B_j(s,d) \geq \tau\}$, with value $+\infty$ when no crossing occurs. The filing date is $f_j = c_j + T$, and disclosure occurs when $a_j = 1$ and $f_j \leq H$. At a filing, the reported stake is $B_j^F = B_j(s, f_j)$ and the remaining order is $Q_j^F = b_j^*(s) - B_j^F$. A finite coarsening Γ maps the stake increment into the pooled order mark $q_{jd}(s) = \Gamma(B_j(s,d) - B_j(s,d-1))$, so observed pooled flow is $X_d = q_{jd} + z_d$.

Market makers observe the pooled history $\mathcal{H}_d^P = (X_0, \ldots, X_d;\ \text{flag landed by } d)$. A filing reports $F = (B^F, a = 1)$, and the flagged tuple adds the residual order, $\mathsf{S}_F = (B^F, Q^F, a = 1)$. At the control node, public information is $\mathcal{H}_H^P$ in the pooled cell and S_F in the flagged cell; the bidder's ξ remains private. I write these cells as $\mathcal{C}_P$ and $\mathcal{C}_F$. They are exclusive and exhaustive. The engagement posterior is $\pi(\mathcal{I}) = \mathbb{P}(a = 1 \mid \mathcal{I})$, and equals one after a filing. The bidder enters with probability

$$p(\mathcal{I}) \;=\; 1 - \Phi\!\left(\frac{P + K + m_0 + \pi\Delta_m - \bar{S}}{\sigma_\xi}\right) \;\in\; (0,1). \tag{1}$$

With B the entry indicator, the terminal shareholder payoff is

$$Y \;=\; (1 - \mathsf{B})\,(v + a\Delta_V) \;+\; \mathsf{B}\,(P + m_0 + a\Delta_m), \tag{2}$$

and prices satisfy the inner fixed point $P(\mathcal{I}) = \mathbb{E}[Y \mid \mathcal{I}]$. Two conventions are part of the model rather than notation. First, $P_{-1}^P := \mathbb{E}[Y]$ is the pre-trading pooled price, which is needed whenever $c = 0$, as $T = H$ forces on every flagged history. Second, $P_{\mathrm{ND}}(\mathcal{H}_{f^-}^P) := P_{f^-}^P$: the not-yet-disclosed price is the last pre-filing pooled price at the *same* realised order flow, whose history already carries "flag not landed by $f-1$", and not a never-disclosed counterfactual. Throughout, c^- and f^- denote the business days immediately before the crossing date and the filing date. The price-path objects are the run-up path $R_d = P_d^P - P_{c^-}^P$, the cumulative run-up $R = P_{f^-}^P - P_{c^-}^P$, and the filing-day jump $J = P^F - P_{\mathrm{ND}}$; R_d, R and J are unsigned, and J is not claimed to be κ-invariant.

The genuine fixed point sits at the control nodes. At an earlier pooled date $d < H$ the price is a tower expectation of already-solved control-node values, with no self-reference; only the control-node map is a fixed point to be solved.

The blockholder evaluates a plan by the expected terminal value of the position it builds, less the trading outlay and the cost of engagement:

$$U_j(s) \;=\; \mathbb{E}\Big[\, b_j^*(s)\, Y \;-\; \mathcal{C}_j^{\mathrm{trade}} \;-\; a_j\, C_j(s) \;\Big|\; s,\, j \Big], \tag{3}$$

where $\mathcal{C}_j^{\mathrm{trade}}$ is the execution outlay, with increments valued at the pooled prices P_d^P up to the plan's last pooled date and $Q_j^F(s)\, P^F$ when $D_j = 1$, and $C_j(s) \geq 0$ is the engagement cost. Only two properties of (3) are ever used: *plan-locality*, that U_j depends on j only through the executed stake path, the prices paid on it, the terminal stake, the engagement flag and the cost; and *integrability*, $\mathbb{E}[\max_j |U_j|] < \infty$ under the finiteness conditions of Section 3.3.

The cost in (3) is not tied to a particular date. It can be booked when the plan is completed or treated as sunk once the filing arrives. These readings give the same comparison among flagged round-two deviations, so the existence argument does not depend on the convention.

The primitive draws are mutually independent, with strictly positive variances. The menu, order-mark support, noise support, and calendar are finite; prices and payoffs are locally bounded on the maintained parameter set; and $\mathbb{E}[\max_{j \in \mathcal{J}} |U_j|] < \infty$ throughout the cutoff set.

Four restrictions on the primitive table travel together, and several results below consume them as a block. Clause (TR-i), *distributional forms and signs*: the distributions above hold as stated, with $\sigma_v^2, \sigma_\varepsilon^2, \sigma_\xi^2 > 0$, $K > 0$, $\bar{z} > 0$, $\Delta_m > 0$, $\Delta_V \geq 0$, and

$$m_0 \geq 0, \tag{4}$$

so that $\bar{m}(\mathcal{I}) = m_0 + \pi(\mathcal{I})\Delta_m \geq 0$. Restriction (4) is what makes the inner pricing root exist, be unique and be continuous; when it is dropped, executed counterexamples produce both no root and three roots. The initial stake satisfies $b_0 < \tau$, so a crossing that predates the model is outside the core analysis. Clause (TR-ii), *stake-path regularity*: Voice stakes are weakly increasing in the calendar date and in the signal, Hold stakes are constant, Exit stakes are weakly decreasing in the date, and every map $s \mapsto B_j(s,d)$ is Borel. Borel regularity is automatic for Voice and for Hold and is a genuine addition for Exit, where the primitives supply monotonicity in the date alone; without it the pooled prices are not defined, because pooled pricing integrates over every type. Stake levels themselves remain continuum-valued: the finiteness above covers the menu, the image of Γ, the noise support and the calendar, not the stake level. Clause (TR-iii), *legal-clock objects*: the crossing, filing, stake-at-filing, residual-order and terminal-target definitions hold as stated, together with $D = 1 \Rightarrow a = 1$, $Q^F \geq 0$ for Voice plans, and, at fixed policies, $T' < T$ implying $B^F(T') \leq B^F(T)$ and $Q^F(T') \geq Q^F(T)$. Clause (TR-iv), *pricing and entry*: the terminal payoff is (2), prices obey $P(\mathcal{I}) = \mathbb{E}[Y \mid \mathcal{I}]$, entry obeys (1), the control-node information set is the fill given above, and the two price conventions $P_{-1}^P = \mathbb{E}[Y]$ and $P_{\mathrm{ND}} = P_{f-}^P$ hold.

3.4 Equilibrium notion

I use a cutoff perfect Bayesian equilibrium, and it carries six requirements: (i) a weakly ordered cutoff vector $k = (k_1, \ldots, k_{J-1})$ selects the plan as a function of the signal; (ii) the pooled and flagged actions are sequentially optimal at every history; (iii) beliefs obey Bayes' rule at every history reached with positive probability; (iv) prices solve their competitive fixed points; (v) the bidder follows (1); and (vi) off-path beliefs are limits along one fixed full-support perturbation family over plans. Weak inequalities allow an action region, including Hold, to collapse. The outer cutoff map is solved on a compact ordered polytope, and uniqueness is not claimed.

The timing and primitive restrictions accompany every result. The cutoff construction requires adjacent plan payoff differences to cross at most once in the signal, with the preferred plan weakly increasing in the signal. It also requires a common compact bracket containing an indifference signal for every adjacent plan pair at every point of Θ, and a continuous, self-mapping outer best-response map $\mathcal{T}(k; \vartheta)$. The inner pricing root is unique and continuous in its belief summaries under (4); continuity of the composed price family in the cutoff vector is the part retained as a condition.

On a flagged history, $(B^F, Q^F, a = 1)$ identifies the informed plan component, so the remaining pooled order is independent noise conditional on that tuple. For the fixed-policy comparisons, the composed terminal target $s \mapsto b_{j(s)}^*(s)$ must be strictly increasing on the flagged signal region for every $k \in \Theta$. The existence construction uses the stronger statement that $(j, s) \mapsto (B_j^F, Q_j^F, a_j)$ is injective over all flagged pairs, including pairs not selected by the cutoff policy.

The pooled-cell comparison uses a strong support restriction. Its posterior law must have the three-point form

$$\mathbb{E}[h] \;=\; A_0(\kappa)\,h(0) \;+\; A_{1/2}(\kappa)\,h(\bar\pi/2) \;+\; A_1(\kappa)\,h(\bar\pi), \tag{5}$$

with $A_0' = A_1' = A_\kappa'$, $A_{1/2}' = -2A_\kappa'$, and a non-positive chord whose absolute value is weakly increasing in $\bar\pi$. Two clauses carry the restriction's remaining bite: (τ-i), *kernel through posterior*, that h depends on the information set through the posterior alone; and (τ-ii), *κ-free support*, that the three support points, the upper point $\bar\pi$ included, do not move with κ. I write the restriction A(τ). Whether the two-round pooled cell satisfies the support condition is open; the implemented calibration does not.

A threshold comparison $\tau' < \tau$ at fixed policies and a common κ needs a bridge between the two pooled classes, written A(br), in five clauses. (br-i), *representation at both policies*: (5) holds under τ and under τ', with endpoints $\bar\pi(\tau)$ and $\bar\pi(\tau')$ and coefficients $A_\kappa'(\tau)$ and $A_\kappa'(\tau')$. (br-ii), *κ-localisation*: all κ-dependence of M_P sits in the weights, so that $\partial_\kappa M_P = \Delta_m A_\kappa' C_h(\bar\pi)$ exactly, with no composition remainder; that last step is derived rather than assumed, and the clause names the same object as (τ-i) against the reading $h = \pi\, p(\hat v, \pi)$ rather than adding an independent restriction. (br-iii), *coefficient stability*: $|A_\kappa'(\tau')| \le |A_\kappa'(\tau)|$, whose weakest sufficient form is equality, reclassification changing which histories are pooled rather than the κ-responsiveness of the pooled weights. (br-iv), *endpoint linkage*: $\bar\pi$ is the chord endpoint of (5) and is the same weakly increasing function of the pooled prior engagement share $\bar\pi_{\mathrm{pr}} = \mathbb{P}(a = 1 \mid D = 0)$ at τ and at τ', the identity branch $\bar\pi = \bar\pi_{\mathrm{pr}}$ being excluded as degenerate. (br-v), *comparability of the chord functional*: $C_h(\cdot)$ and the kernel behind it are the same functions of the posterior at both thresholds; without it the threshold comparison sets $|C_h|$ against a different functional and means nothing. These clauses are used only for the threshold sensitivity result. For the general-equilibrium result I require a twice continuously differentiable outer map on a candidate region $\mathcal{R}$, a contraction bound $L_{\mathcal{R}} < 1$, and a constant sign for the equilibrium liquidity derivative.

3.5 The cutoff polytope and the premium objects

The cutoff vector is $k = (k_1, \ldots, k_{J-1})$ with weak ordering. The compact, convex set of such vectors is Θ, and ϑ collects the remaining parameters. The outer map $\mathcal{T}(k;\vartheta)$ gives the cutoff vector implied by a conjectured vector k. On a candidate region $\mathcal{R}$, write $L_{\mathcal{R}} = \sup_{\mathcal{R}} \|D_k\mathcal{T}\|$ and use $r_\tau = -\tau$ and $r_T = -T$ so that a larger r means a tighter rule. The fixed-policy attenuation margin is

$$g_r^{PE} \;=\; -\,\mathrm{sgn}\big(d\Delta^{\mathrm{act}}/d\kappa\big)\;\partial_{\kappa r}\Delta^{\mathrm{act}}, \tag{6}$$

and the inversion-free cutoff bounds are $\bar k_x = |\partial_x\mathcal{T}|/(1 - L_{\mathcal{R}})$ and $\bar k_{\kappa r}$. The general-equilibrium remainder is

$$\mathcal{B}_r^{GE} \;=\; |\Delta_{\kappa k}|\,\bar k_r \;+ \big(|\Delta_{kr}| + |\Delta_{kk}|\,\bar k_r\big)\bar k_\kappa \;+ |\Delta_k|\,\bar k_{\kappa r}. \tag{7}$$

Here the Δ terms are the gradient and cross-partials of Δ^{act} in (k, κ, r) along the equilibrium branch. The dominance-and-contraction region $\mathcal{R}_r$ has slack $\eta_r = g_r^{PE} - \mathcal{B}_r^{GE}$ and may be empty.

The engagement component of the premium is built from $h(\mathcal{I}) = \pi(\mathcal{I})p(\mathcal{I})$. The kernel satisfies $h \ge 0$ and $h(0) = 0$, and the expected engagement-related premium is

$$\Delta^{\mathrm{act}} \;=\; \Delta_m\,\mathbb{E}\big[h(\mathcal{I}_H)\big] \;\ge\; 0. \tag{8}$$

The cell averages are $M_F = \Delta_m \mathbb{E}[h \mid D=1]$ and $M_P = \Delta_m \mathbb{E}[h \mid D=0]$ when the corresponding cell has positive mass. The flagged weight is $\Omega = \mathbb{P}(D=1) = \mathbb{P}(a=1)\omega_a$, where $\omega_a = \mathbb{P}(D=1 \mid a=1)$. In the three-point representation, $\bar{\pi}$ is the upper support point, not the mean pooled engagement share. The latter is κ-invariant under the support restriction and is strictly below $\bar{\pi}$ away from the degenerate case; it equals $\bar{\pi}/2$ only when $A_0 = A_1$. Define $\mathcal{S} = |\partial_\kappa \Delta^{\text{act}}|$ and $\mathcal{S}_P = |\partial_\kappa M_P|$. The chord is

$$C_h(\bar{\pi}) \;=\; h(0) \;-\; 2h(\bar{\pi}/2) \;+\; h(\bar{\pi}), \tag{9}$$

maintained non-positive with $|C_h|$ weakly increasing in $\bar{\pi}$, and A'_κ is the common derivative of the weights in (5), bounded on $[0,1]$. The weight-effect ratios are W_τ and W_T, for instance $W_T = (1-\Omega(\tau,5))/(1-\Omega(\tau,10))$, which is at most 1 when Ω rises. The composition-effect ratios are C_τ and C_T, for instance $C_T = \mathcal{S}_P(\tau,5)/\mathcal{S}_P(\tau,10)$, which are unsigned. The margin subscript on a composition ratio is always written, because C is otherwise overloaded by the chord C_h, the engagement cost $C_j(s)$ and the cells $\mathcal{C}_F, \mathcal{C}_P$.

Reading $\bar{\pi}$ as the mean of the pooled posterior law rather than as its upper support point forces a point mass at $\bar{\pi}$ with $A'_\kappa = 0$ and zero interior motion for every kernel. That is degenerate, and it is excluded throughout.

4 Equilibrium existence

Existence is standing ground here, not the claim. The claim the paper is making is the attenuation result of Section 5; what this section supplies is the conditional on which the whole construction rests.

Under the hypotheses (P-1)–(P-13) below, a cutoff perfect Bayesian equilibrium over complete contingent plans exists at *every* $\kappa \in [0,1]$: there is $k^\star \in \Theta$ with $k^\star = \mathcal{T}(k^\star; \vartheta)$, prices at their inner fixed points, and beliefs Bayes-consistent on path. Uniqueness is not claimed, in any form.

Five things travel with that conclusion. Off-path beliefs are limits of *one* full-support perturbation family over plans, fixed once and used to define the price system at every $k \in \Theta$ and not only at $k^\star$, because the deviation payoffs that define $\mathcal{T}$ read off-path pooled histories. They are supplied at every pooled history reachable with positive probability under some plan profile. The boundary values of κ are handled by *extension* rather than by restriction: at $\kappa \in \{0,1\}$ the noise support degenerates to $\{0\}$ and to $\{\pm\bar{z}\}$, and a pooled history needing a mark outside the surviving support is null under every profile, so it lies off nature's path rather than off the players', carries no requirement under clause (vi) of the equilibrium definition, and is read by no step. No cut to $\kappa \in [0,1)$ is taken. Flagged-tuple beliefs are the point mass at the unique generating pair, supplied at every tuple in the image of $(j,s) \mapsto (B_j^F, Q_j^F, a_j)$, on path and off, since that image includes tuples the cutoff vector does not select; the point mass is a *version* of the conditional law at every image tuple, which is what a conditional law is when the signal is continuous, and any a.e.-equal version serves clauses (iii) and (vi) equally. No tuple outside the image arises, because the round-two action-set hypothesis (P-9) leaves no off-menu order to produce one. The bidder follows (1). And there is a sequentially optimal flagged component at *every* flagged pair (j,s), selected or not: the flagged price is invariant across the round-two deviation set, so the flagged order cancels out of the continuation and (P-10) makes what remains constant.

The hypotheses. The list is long, and every clause of it is load-bearing, so it is enumerated rather than gathered into a sentence.

(P-1) Independent primitives.

(P-2) The finite model and integrable objective.

(P-3) Ordered plans and single crossing.

(P-4) Legal-clock discipline.

(P-5) The compact continuous outer self-map, read as set out below.

(P-6) *Joint tuple injectivity*: the map $(j, s) \mapsto (B_j^F, Q_j^F, a_j)$ is injective on the whole flagged-pair set $\{(j, s) : D_j = 1\}$, *including pairs no cutoff vector selects.* This is strictly stronger than the on-path form, and it is the form the argument consumes where it pins *off-path* flagged beliefs.[1]

(P-7) The disclosure partition and price-path decomposition of Section 5.1, by statement, *with its own hypotheses travelling.*

(P-8) The no-feedback timing, read together with the flag-termination clause.

(P-9) The *definitional* round-two action-set hypothesis: the flagged-round action set *is* the plan-generated set $\{Q_{j'}^F(s)\}$ taken over menu elements agreeing with j on everything already played. This is not a closure condition; the closure form is jointly unsatisfiable with finiteness, by cardinality.

(P-10) *Continuation-cost equivalence on that same set*: menu elements sharing j's pooled path up to $f_j(s)$ and carrying $a_{j'} = a_j$ have the same engagement cost, $C_{j'}(s) = C_j(s)$. It holds trivially on any single-Voice menu, where the set is a singleton, and is live only on menus with two or more Voice plans sharing a pooled path. What it buys, under the plan-completion reading of the cost timing, is this: on that set the flagged price does not move and the flagged order cancels, so the engagement cost is the only thing that can differ between staying and deviating, and at a flagged pair the cutoff vector does not select there is no date-0 optimality to fall back on, so without the clause the deviator takes the class member with the smallest cost and clause (ii) of the equilibrium definition fails at that node. Under the sunk reading the continuation is constant on the deviation set with no clause at all and the hypothesis is not consumed; it is listed because the statement does not commit to a reading, and it is what makes the conclusion hold under both.

(P-11) The restriction $m_0 \geq 0$ of (4).

(P-12) The blockholder's objective (3), whose $-a_j C_j(s)$ term carries the engagement cost, together with the cost timing convention: the cost may be booked on completing the plan or treated as sunk once the filing has landed, the two give the same round-two comparison on the round-two deviation set, and the result does not depend on the choice.

[1]The distinction is not cosmetic. An earlier recorded statement of this result carried the on-path form while the proof consumed the joint form. The joint form is stated here because the proof uses it to pin off-path flagged beliefs; the on-path form alone would not suffice. Which form is meant is named every time either appears.

(P-13) The primitive table restrictions the argument consumes, in particular: the terminal payoff (2) with the price convention $P(\mathcal{I}) = \mathbb{E}[Y \mid \mathcal{I}]$ and the entry rule (1), both from clause (TR-iv); clause (TR-ii)'s Borel regularity for *every* plan including Exit, needed here *directly* and not by way of the partition result, whose conclusion is measurability of D and of the cell map; clause (TR-iii)'s implication $D = 1 \Rightarrow a = 1$, the definitions of c, f, B^F, Q^F and b^*, and $\partial_s B_j \geq 0$ for Voice; and clause (TR-i)'s distributional forms with $\Delta_m > 0$.

Two readings. Both belong to the statement rather than to the proof. First, the pricing assumption of Section 3.4 *is not assumed here*: its existence and uniqueness content is derived from $m_0 \geq 0$, its continuity content *in the belief summaries* $(\hat{v}, \pi)$ from the same scalar reduction, and its measurable-selection content from joint tuple injectivity together with clause (TR-ii). What is *not* derived is its cutoff clause, the continuity of the composed pooled price family in the cutoff vector k, which runs through the conditioning pair $(\hat{v}, \pi)$ rather than through the pricing map. That continuity enters only through the second reading, and Section 5.6 records it measured to fail at the implemented calibration. Second, the outer-map assumption *is read* as asserting that $\mathcal{T}$, under a named tie-break-and-corner selection, without which a correspondence cannot be called continuous, is a well-defined, single-valued, continuous self-map of the nonempty polytope Θ; and, in its *bracket* half, that there is a fixed compact signal interval, **the common bracket** $[s_{\mathrm{lo}}, s_{\mathrm{hi}}]$ at the given ϑ, common in $k \in \Theta$ and in the adjacent plan pair, of which Θ is the ordered part ($s_{\mathrm{lo}} \leq k_1 \leq \cdots \leq k_{J-1} \leq s_{\mathrm{hi}}$), such that for every $k \in \Theta$ and every adjacent plan pair the best-response indifference signal **exists** and lies in the bracket. That is what makes the finite cutoff vector represent the best-response plan map on the whole Gaussian signal line and not merely inside the bracket. The clause excludes every menu-and-price-system at which some adjacent indifference signal fails to exist, never-selected middle plans as well as dominated extreme plans; the stronger form is taken so that the cutoff vector carries interior data at every k, and the exclusion is a named domain narrowing.

Finally, at any such equilibrium at which the interior-crossing condition holds, both cells carry strictly positive probability and both are on path; restating that condition as a single signal threshold needs two further hypotheses, Voice stake monotonicity across plans and engagement flags equal to one *exactly* on an upper set of the ordered menu.

How the proof goes. The argument builds the equilibrium object rather than characterising it, and the online appendix carries it in full. A conjectured cutoff vector induces a measurable plan-selection map, under which every date-0 object is a deterministic Borel function of the pair (j, s). The inner price system at that conjecture separates a finite pooled layer from a continuum-indexed flagged layer: the scalar reduction behind (4) gives existence, uniqueness and continuity of the inner root, and the flagged layer is priced by inverting the flagged-pair map, which joint injectivity makes invertible. Beliefs are then supplied at every history that carries a requirement, Bayes where the reaching weight is positive and the fixed perturbation family's limit elsewhere, and sequential optimality of the flagged component is discharged at every flagged pair, selected or not. The outer best-response map is built on the common bracket; that it is weakly ordered and carries Θ into itself follows from single crossing, while the bracket and the continuity of the map are taken rather than derived. Brouwer's fixed-point theorem then applies, and the six requirements of the equilibrium definition are assembled at the fixed point, with the boundary values of κ handled by extension. The proof closes with six disclaimers: uniqueness in no form; the interior-crossing addendum conditional at the equilibrium and nowhere else; boundary extension, which is where a withdrawn claim of a belief at every pooled history at $\kappa = 1$ sat; one fixed convention; flagged-order optimality without uniqueness; and two hypotheses that are sufficient rather than necessary and are not verified menu by menu.

5 Liquidity and control outcomes

This section states what the model delivers about the control outcome. The order is the order the model builds it in: the partition and its accounting identities, then what liquidity does inside each cell, then the attenuation theorem, then what survives the move from fixed policies to equilibrium. Section 5.6 then reads the whole stack against the implemented calibration and says plainly what it makes of the antecedents. Each statement carries its conditionality with it; the full numerical records sit in the online appendix.

5.1 The partition and its accounting identities

Three facts about the partition come first, and the rest of the section rests on them. They hold under the independence, finiteness, legal-clock and pricing conditions of Section 3, the cutoff selection map of clause (i) of the equilibrium definition, and the two price conventions $P^P_{-1} = \mathbb{E}[Y]$ and $P_{\mathrm{ND}} = P^P_{f^-}$; part (c) also consumes clause (TR-ii)'s Borel regularity of $s \mapsto B_j(s,d)$ for *every* plan including Exit, because pooled pricing integrates over every type, and a content for the control-node information set. (a) The disclosure indicator $D = \mathbf{1}\{a = 1,\ c(\tau) + T \le H\}$ is measurable, and it maps every control-node history into exactly one cell. (b) For every Voice plan, $f_j \le H$ if and only if $B_j(s, H - T) \ge \tau$. (c) Each flagged history yields B^F, R_d, R and J, and these satisfy the exact identity

$$P^F - P^P_{c^-} \;=\; R + J. \tag{10}$$

Part (b) is the *clock equivalence*, and the threshold and window results both consume it: it converts a statement about the filing date into a statement about the stake path T days before the horizon, which is what makes window comparisons tractable. Identity (10) says that the total price move from just before the crossing to the flagged price decomposes exactly into the cumulative run-up plus the filing-day jump.

The premium then decomposes across the two cells. Given the partition, the definitions of Section 3.3, the version of $\mathbb{E}[Y \mid \mathcal{I}]$ pinned by the pricing assumption, finiteness of Δ_m, and one common probability space, whenever $0 < \Omega < 1$,

$$\Delta^{\mathrm{act}} \;=\; \Omega\, M_F \;+\; (1-\Omega)\, M_P. \tag{11}$$

At $\Omega = 1$ this degenerates to $\Delta^{\mathrm{act}} = M_F$ and at $\Omega = 0$ to $\Delta^{\mathrm{act}} = M_P$; in each degenerate case the average over the null cell is *undefined rather than imputed.* That last clause is a non-identification statement, and it is proved rather than asserted.

5.2 Liquidity in the flagged cell

Liquidity does nothing inside the flagged cell. At fixed cutoff *and* execution policies, under the same standing conditions together with the on-path injective identification condition consumed almost surely on the flagged set,[2] the partition result above, the no-feedback timing, $\Omega > 0$, and an explicit bidder-entry rule, either (1) or any rule with the two properties named in the proof and carried as bookkeeping, the flagged tuple $\mathsf{S}_F = (B^F, Q^F, a{=}1)$ makes the pre-filing pooled history conditionally independent of (v, s, ξ) on the flagged set. The flagged posterior, the flagged price, the entry probability and M_F are therefore all invariant to κ.

The weak identification wording of the identification assumption is not sufficient here: it permits two pairs (j, s) with different pooled paths, which is this result's first failure case. The form consumed is the on-path injective one, named as such. This is the theoretical statement behind the empirics' identification claim: the flagged half of the timing split should not move with liquidity, and the pooled half should carry the entire liquidity derivative.

5.3 Liquidity in the pooled cell

Inside the pooled cell liquidity does move the premium, but only under the support restriction $\mathrm{A}(\tau)$, and the restriction is the substance of the result rather than a caveat on it. Assume $\mathrm{A}(\tau)$, both clauses (τ-i) and (τ-ii); $h(0) = 0$; κ-free pooled mass and κ-free pooled engagement moment at fixed policies; the partition result by statement; and regularity stated minimally, h continuous on $[0, \bar{\pi}]$ and twice differentiable on $(0, \bar{\pi})$, with Darboux's theorem doing the rest and no continuity of h'' required. Then (a) $\partial_\kappa \mathbb{E}_\kappa[h] = A'_\kappa\, C_h(\bar{\pi})$, exactly; and (b) $C_h(\bar{\pi}) = \frac{1}{4}\bar{\pi}^2 h''(\zeta)$ for some $\zeta \in (0, \bar{\pi})$, an identity rather than an approximation. Adding a second-order Peano expansion of h at $0+$ and one and the same kernel along a shrinking family gives (c) $C_h = \frac{1}{4}h''(0)\,\bar{\pi}^2 + o(\bar{\pi}^2)$, so the interior motion vanishes at rate $\bar{\pi}^2$ as $\bar{\pi} \downarrow 0$; the seam at which the threshold result consumes part (c) needs $|A'_\kappa|$ bounded *uniformly in* $\bar{\pi}$ along the limit. The statement is an "if" and never an "iff": $A'_\kappa = 0$ also kills the interior motion.

Whether the two-round pooled cell of Section 3.2 satisfies the support condition of $\mathrm{A}(\tau)$ is **open**: the online appendix shows that the support condition is the assumption's entire remaining bite, and exhibits a one-round market that satisfies it and a no-disclosure structure that does not.

Tightening the threshold then moves the pooled cell in three steps. Take $b_0 < \tau' < \tau$ at a common window T and a common κ, with $\Omega(\tau', T) < 1$, at fixed policies. Leg 1, which is unconditional: $\mathcal{C}_F(\tau, T) \subseteq \mathcal{C}_F(\tau', T)$, with every newly flagged history generated by a Voice plan, so $\Omega(\tau', T) \geq \Omega(\tau, T)$. Leg 2, also unconditional: the pooled engagement *share* falls, $\bar{\pi}_{\mathrm{pr}}(\tau') \leq \bar{\pi}_{\mathrm{pr}}(\tau)$, with an exact identity for the difference between the two shares. Leg 3, under $\mathrm{A}(\mathrm{br})$: $\mathcal{S}_P(\tau', T) \leq \mathcal{S}_P(\tau, T)$, with equality whenever $C_h(\bar{\pi}(\tau)) = 0$. Legs 1 and 2 need only the clock equivalence, the no-feedback timing, fixed policies, $b_0 < \tau' < \tau$ imposed at *both* thresholds, independence, legal-clock discipline, clause (TR-iii)'s implication $D = 1 \Rightarrow a = 1$, and $\Omega(\tau') < 1$; nestedness is a *conclusion* of leg 1, not a hypothesis of it. Leg 3 also needs the pooled interior-motion result by statement, the maintained *magnitude* monotonicity of $|C_h|$ in $\bar{\pi}$ from $\mathrm{A}(\tau)$, the sign half $C_h \leq 0$ being unused at this leg, and clauses (br-i)–(br-v) of $\mathrm{A}(\mathrm{br})$.

[2] "Almost surely" is the permissive reading, and it is the only coherent one when B^F is continuum-valued, since then no individual flagged tuple has positive probability.

Leg 3 inherits the conditionality of A(br), and through clause (br-i) it inherits the representation (5) at *both* compared thresholds; the numerical record in the online appendix therefore bears on leg 3 as directly as it bears on the interior-motion result. At the implemented calibration leg 3's antecedent is not satisfied, and at that calibration leg 3 says nothing about the implemented pooled cell. Legs 1 and 2 are unaffected: they consume no chord restriction. Clause (br-iii) is the clause with the least justification behind it and is the one to attack first.

5.4 Disclosure attenuation

Theorem 1 (Disclosure attenuation at fixed policies)**.** *At fixed plan and cutoff policies, with* $0 < \Omega < 1$ *and* $\mathcal{S}_P > 0$*, and under the hypotheses (T-1)–(T-15) listed below:*

(A) Factorisation. $\mathcal{S} = (1-\Omega)\,\mathcal{S}_P$*, exactly; and the same factorisation holds for the total-variation aggregate of* Δ^{act} *over any* κ*-grid, with no differentiability required.*

(B) Threshold margin. *Threshold tightening attenuates:*

$$\frac{\mathcal{S}(\tau')}{\mathcal{S}(\tau)} \;=\; W_\tau\, C_\tau \;\le\; 1, \tag{12}$$

because both *ratios lie in* $[0,1]$*. No dominance condition is needed.*

(C) Window margin. *Window tightening attenuates* if and only if $W_T C_T \le 1$*, where* $W_T \le 1$ *is proved — from the clock equivalence of Section 5.1 and the monotone Voice stake path — and* C_T *is* unsigned*. The equivalent form*

$$\frac{\partial_{r_T}\mathcal{S}_P}{\mathcal{S}_P} \;\le\; \frac{\partial_{r_T}\Omega}{1-\Omega} \tag{13}$$

holds on average along the tightening path*, integrated over* $[-T,-T']$*, and exactly in the infinitesimal limit; read pointwise,* (13) *is false.*

The hypotheses are: (T-1) fixed policies; (T-2) the interior-crossing condition at each compared policy; (T-3) $\mathcal{S}_P > 0$*; (T-4) the two-cell decomposition* (11)*; (T-5) the flagged-cell invariance of Section 5.2, with its own hypotheses travelling; (T-6) the partition result of Section 5.1; (T-7) PE-*Ω*, that is* $\partial_\kappa\Omega = 0$ *at fixed policies — derivable rather than assumed, and it is exactly what fails in general equilibrium, which is the term Proposition 2 bounds; (T-8)* κ*-differentiability of* M_P*, which no standing hypothesis of Section 3 supplies and which is therefore carried in the proof; (T-9) the support restriction* $\mathrm{A}(\tau)$ *at* both *compared policies, with* $\overline{\pi}$ *read as the upper support point; (T-10) the pooled interior-motion result of Section 5.3; (T-11) clauses (br-i)–(br-v) of* A(br) *at the threshold pair; (T-12) the three-leg threshold result of Section 5.3; (T-13) the no-feedback timing; (T-14) a smooth window interpolation, for the local form* (13) *only; and (T-15) threshold-side smoothness, which has been confirmed non-load-bearing.* No unconditional window sign is claimed.

Part (A) is the weight effect in its exact form: the liquidity sensitivity of the expected engagement-related premium is the pooled cell's sensitivity, scaled by the pooled cell's weight. Part (B) is the threshold-margin result and needs no dominance condition, because the weight ratio and the composition ratio both lie in $[0,1]$. Part (C) is the window-margin result, and it is an "if and only if" rather than a sign: the composition ratio C_T is unsigned, so the model does not deliver window-margin attenuation as a theorem. The two margins of the disclosure rule therefore enter the policy discussion with different epistemic status, and the February 2024 acceleration, a window change, lands on the margin where the theory signs nothing.

The disclosure-regime comparison, and why it is not a window test. A separate numerical comparison is sometimes read as evidence about the window, and it is not one. In the one-round predecessor model the expected engagement-related premium can be computed twice: once with the market seeing the disclosure flag alongside order flow, once with order flow alone. The cutoffs k_1 and k_0 stay at the baseline equilibrium throughout and only the disclosure cutoff k_D moves; moving it across four values generates the flagged weights $\Omega = 0.037$, 0.129, 0.286 and 0.50. The ratio of liquidity sensitivities (flagged divided by pooled, total variation over $\kappa \in [0.15, 0.85]$) is 1.064, 1.184 and 1.136 at the first three of those weights, falling to 0.378 at $\Omega = 0.50$, with the sign crossing located by bisection at $\Omega^\star = 0.343$. Above one, the flag makes premia *more* liquidity-sensitive. Figure 1 plots the two curves at the baseline. This is a *disclosure-regime* comparison (the flag observed versus hidden, at a fixed filing window), and its ratios are regime-comparison composition outcomes: they are not $W_T C_T$, they measure no window pair, and the predecessor model has no window to vary. The comparison earns its place for one reason: it shows that a composition factor can exceed one, which is what motivates the genuine window-margin "if and only if" of Part (C) and warns against reading attenuation into any flag comparison.

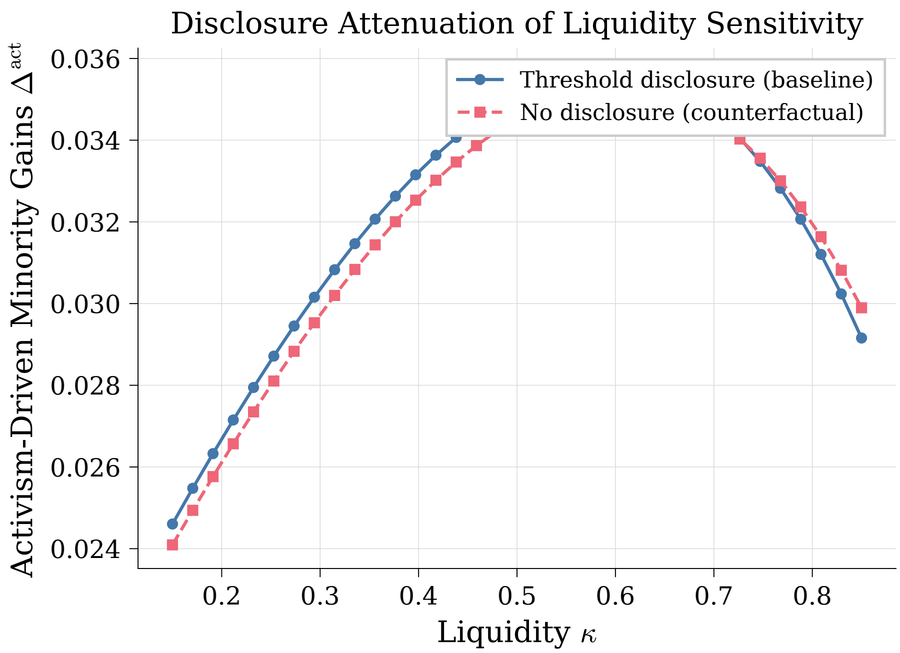

Figure 1: The disclosure-regime comparison in the one-round predecessor model, at fixed cutoffs ($\Omega = 0.037$). The solid line with circle markers prices the blockholder's engagement-driven premium with the market seeing order flow and the disclosure flag (threshold disclosure, baseline); the dashed line with square markers prices it from order flow alone (no disclosure, counterfactual). The two curves cross near $\kappa = 0.73$. At this baseline the flagged cell is *more* liquidity-sensitive than the pooled one (total variation ratio 1.064), so this comparison refutes attenuation in κ for the regime margin at this weight. It is a composition fact, not a window test; the window record is Table 1.

5.5 From fixed policies to equilibrium

Theorem 1 holds at fixed policies, and hypothesis (T-7) is exactly what fails once the cutoff vector is allowed to move with κ. The next result asks what survives that move. It carries the region as a *named hypothesis* rather than exhibiting one.

Proposition 2 (The dominance-and-contraction implication in general equilibrium)**.** *Assume:*

(C-1) the along-the-path contraction $L_{\mathcal{R}} < 1$ *of Section 3.4, in* one fixed norm convention *— an induced operator norm with its dual pairings; a mismatched pairing silently voids the implication;*

(C-2) $\mathcal{R}_r$ *relatively open in* both *coordinates, with* $\kappa \notin \{0, 1\}$*;*

(C-3) an interior single-branch equilibrium;

(C-4) twice continuous differentiability of Δ^{act} *in* (k, κ, r)*;*

(C-5) a non-vanishing equilibrium *liquidity derivative — the equilibrium sensitivity* $\mathcal{S}^{GE}$*, which is distinguished from the fixed-policy sensitivity* $\mathcal{S}$ *of Section 3.5;*

(C-6) strict dominance $g_r^{PE} > \mathcal{B}_r^{GE}$ *on the region, with* g_r^{PE} *as in* (6) *and* $\mathcal{B}_r^{GE}$ *as in* (7)*;*

(C-7) the threshold *margin r_T only: the window coordinate is an integer, and nothing local is claimed there.*

Then the fixed-policy attenuation sign survives in equilibrium on the region:

$$\partial_r \mathcal{S}^{GE} \;\leq\; -\eta_r \;<\; 0. \tag{14}$$

The sign-coherence hypothesis — that the fixed-policy attenuation sign agrees with the equilibrium sign — is confirmed unused in (14)*: its work is one of naming, fixing when that conclusion may be called a survival of the fixed-policy sign. The sign-constancy clause of the general-equilibrium condition is likewise unused.*

5.6 The model at the implemented calibration

The results above are conditionals, and this subsection says plainly what the implemented calibration makes of their antecedents. The online appendix carries the full measurements.

Existence. The two main hypotheses about the equilibrium object are measured to fail at the calibration the implementation runs. The single-crossing clause fails at two independently found loci: at one, an adjacent-plan payoff difference crosses zero three times on an open set of the second cutoff; at the other, the preferred plan decreases in the signal across a cell edge. The outer map's continuity fails for the polytope the proof constructs, with measured jumps of 6.33×10^{-3} to 2.83×10^{-2} across steps in the cutoff of at most 2×10^{-9} at the baseline node, and jumps reaching 0.16 at a second node. In plain words: the proof needs the outer best-response map to be continuous and single-valued, and at the implemented calibration that map has measured jumps and stretches where no weakly increasing selection exists at all. Nonexistence is neither claimed nor shown: twenty-three of twenty-seven equilibrium-sweep nodes converge, the remaining four stay unresolved after thirty seeds each, and a discontinuous self-map may still have fixed points.

The chord restriction. The restriction behind the pooled-cell results fails at the same calibration. The pooled cell's engagement-posterior law, enumerated exactly over all 4^{H+1} order-flow paths at $H = 10$, at two hundred nodes, carries between 23 and 767 distinct posterior values where the restriction's representation asks for three, at every one of the 180 non-degenerate nodes; the remaining twenty nodes are degenerate and decide nothing. The interior motion of the pooled premium, leg 3 of the threshold-nesting result and Part (B) of the attenuation theorem remain proved as conditionals; at this calibration they say nothing about the implemented pooled cell. The open question concerns the restriction's domain: a different menu, a different horizon, or a different calibration could still satisfy the support condition.

The window record. At frozen policies, with cutoffs pinned at the baseline equilibrium, the flagged weight is $\Omega = 13.84$ percent, the disclosed share of engagements is $\omega_a = 61.15$ percent, and the cell averages are $M_F = 0.553$ and $M_P = 0.223$ premium percentage points. Cutting the window from $T = 10$ to $T = 5$ then cuts the premium's liquidity sensitivity to between 18 and 77 percent of its long-window value at every checked node, and the composition leg carries almost all of it: C_T runs from 0.21 to 0.79 while the weight leg shaves 3 to 14 percent (Table 1). The same record confirms the flagged cell's invariance exactly, with every flagged object invariant in κ to machine zero over the grid while the pooled M_P moves 0.0722 premium percentage points, and gives the threshold margin the same direction, with zero of eight compared pairs above one. Two caveats travel with the record. At $H = 10$ the long window is the corner $T = H$; the $H = 12$ column of the table, where $T = 10$ is strictly interior, points the same way but travels the chord closed form whose support condition fails at this calibration, so it is directional corroboration rather than a second independent magnitude. And the record is node evidence at one calibration, not a sign theorem: Part (C) claims none. At this calibration the model's directional prediction for the February 2024 acceleration is attenuation; the empirical layer does not test that direction, and the filing-delay evidence of Section 6 speaks to timing only.

Table 1: The window-margin record at fixed policies: cutting the window from $T = 10$ to $T = 5$ at the implemented calibration. Weight leg W_T, composition leg C_T, product $W_T C_T \leq 1$ (attenuation), with no τ node above one. Verified on the numerical grid; sensitivity measured as the total variation of Δ^{act} over the κ grid. The $H = 12$ column re-runs the comparison with $T = 10$ strictly interior to the horizon, where the 0.9-quantile node returns 1.0000 because the τ ladder stops biting and the two flagged weights coincide, not because the product crosses one.

τ quantile	W_T	C_T	$W_T C_T$	$W_T C_T$ at $H = 12$
0.1	0.8559	0.2124	**0.1818**	0.1099
0.3	0.8559	0.2124	**0.1818**	0.1099
0.5	0.8622	0.2384	**0.2055**	0.2406
0.7	0.9176	0.4686	**0.4299**	0.5772
0.9	0.9730	0.7939	**0.7724**	1.0000

The equilibrium region. For the move from fixed policies to equilibrium, eighteen of eighty grid nodes satisfy the two pointwise inequalities, the contraction bound below one and positive dominance slack, with the largest contiguous block, nine nodes, at $T = 5$. A node is a pointwise numerical fact, not evidence of a region: whether any parameter vector satisfies the full antecedent of Proposition 2 is open.

6 Did the clock move? Filing-delay evidence

The empirical layer of the paper is one exercise, and it is descriptive. E1 asks whether the realised delay between the trigger event and the Schedule 13D filing fell after the window margin moved from ten calendar days to five business days on 2024-02-05. It is a before-and-after comparison with no control group: it identifies no effect on liquidity, returns, activism, bidder entry, takeover premia, or control outcomes, and it tests neither the invariance result nor the attenuation theorem. What it establishes is narrower and, for a paper that treats the statutory clock as a primitive, load-bearing: the clock moved realised filing timing. The specification was registered before the run, and every number in this section is drawn from the run's result record.

6.1 Protocol and population

The population is every `SC 13D` or `SCHEDULE 13D` original listed in the EDGAR quarterly form indexes for two windows: 2023Q2–Q3, ahead of the rule's adoption, and 2024Q3–Q4, after the new deadline took effect. Both EDGAR spellings are taken and both amendment spellings excluded;

the enumeration collapses on accession, giving 616 filings pre and 521 post — the near-census of the two windows, not a sample. The unit of analysis is the campaign: one row per subject firm and trigger date, keeping the earliest accession where a reporting group files simultaneously. That leaves 461 eligible units pre and 435 post, of which 450 and 432 resolve to a measured delay; 14 rows (11 pre, 3 post) remain unresolved and are carried through the bounds below.

The trigger date is the filing's own "Date of Event Which Requires Filing," read from the structured XML tag where it exists and from the cover-page text otherwise. The filing date is the EDGAR acceptance timestamp, with acceptance after 17:30 Eastern rolled to the next business day. The delay is counted in federal business days. Inference on the headline difference is a cluster bootstrap on the subject firm, 2,000 draws at seed 20260901, percentile interval at 95 percent. Because the unresolved rows carry no measured delay, every estimate is reported complete-case and bounded, with every unresolved row assigned first to the extreme that most hurts the claim and then to the one that most helps it. Three gates were registered as binding. G1 requires the worst-case lower bound on the post-minus-pre difference to clear zero. G2 caps the gap between the pre and post unresolved shares at ten percentage points — the live hazard, because structured XML became mandatory on 2024-12-18, inside the post window, so post filings parse more easily than pre filings by construction. G3 is a blind hand-audit of the parser. All three pass; the audit details sit with the denominator discussion below.

6.2 The delay distribution before and after the acceleration

Table 2 reports the result; Figure 2 draws the distributions.

Table 2: Realised trigger-to-filing delay of initial Schedule 13D filings before and after the February 2024 acceleration. Enumerated: unique accessions listed as `SC 13D` or `SCHEDULE 13D` originals in the EDGAR quarterly form indexes for the window. Eligible: one campaign per (subject firm, trigger date), earliest accession kept, plus the unresolved rows. Resolved campaigns carry a measured delay: the trigger date is the filing's "Date of Event Which Requires Filing" (structured XML tag first, cover-page text second), the filing date is the EDGAR acceptance timestamp rolled to the next business day when acceptance falls after 17:30 ET, and the delay is counted in federal business days. The share is the fraction of resolved campaigns filed within five business days, complete-case. Bracketed values are worst-case bounds that assign every unresolved row to one extreme and then the other. The difference is post minus pre; its interval is a cluster bootstrap on the subject firm, 2,000 draws, seed 20260901. Descriptive; no control group; no causal claim.

	Pre (2023Q2–Q3) (10-calendar-day rule)	Post (2024Q3–Q4) (5-business-day rule)
Enumerated filings	616	521
Eligible units	461	435
Resolved campaigns	450	432
Unresolved	11	3
Median delay, business days	6 [6, 7]	5 [5, 5]
Share filed within 5 business days	38.9%	67.1%
worst-case bound	[38.0, 40.3]	[66.7, 67.4]
Difference, percentage points	+28.2	
95% bootstrap interval	[+21.8, +34.8]	
worst-case bound	[+26.3, +29.4]	

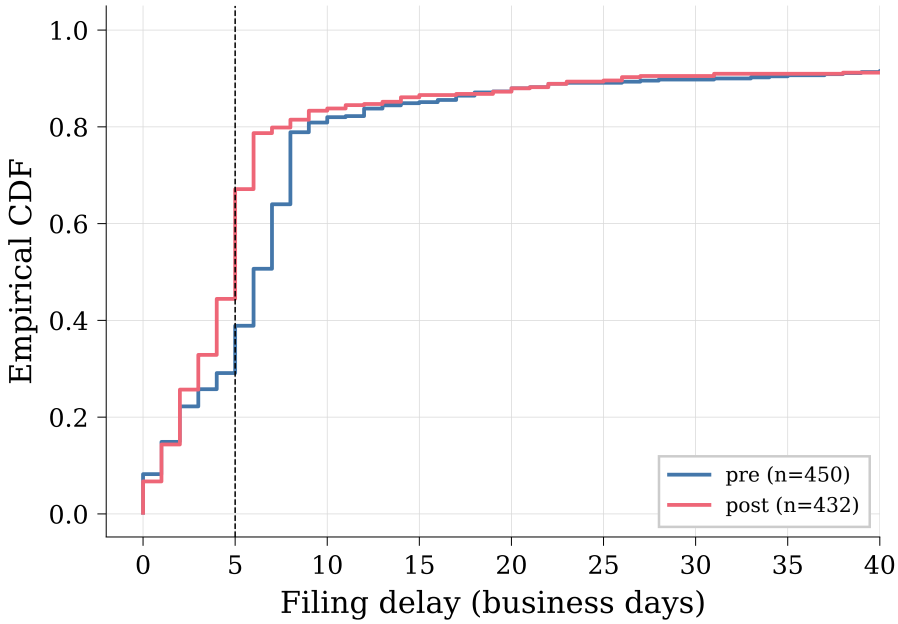

Figure 2: The empirical CDF of the trigger-to-filing delay in federal business days, pre (450 campaigns, 2023Q2–Q3) against post (432 campaigns, 2024Q3–Q4). The dashed line marks the five-business-day deadline in force from 2024-02-05. Resolved campaigns only; there is no outcome screen, so late and catch-up filings stay in the population and form the right tail visible in both periods.

Two readings of the result, each bounded. First, the compression is real and large. The median delay falls from six business days to five, and the worst-case bounds on the median do not overlap — $[6, 7]$ pre against $[5, 5]$ post — so the decline survives any assignment of the unresolved rows. The share filed within five business days rises from 38.9 to 67.1 percent, a difference of +28.2 percentage points with a 95 percent bootstrap interval of $[+21.8, +34.8]$. Nor does the result rest on the unresolved rows: assigning all fourteen to the extremes bounds the difference between +26.3 and +29.4 points, and gate G1, which requires that lower bound to clear zero, passes.

Second, the numbers agree in kind with the Commission's own arithmetic: the release put about 29 percent of 2022's initial filings already inside the amended deadline, against 38.9 percent here in the pre window, and expected earlier filing for about 59 percent of timely 13D reports — different samples and windows, the same direction (U.S. Securities and Exchange Commission, 2023). The population the deadline bit was heterogeneous in exactly the dimension the rule targets: in Bebchuk et al. (2013)'s sample of filings by engaging investors, the mass rises into the deadline bucket, with 45.3 percent of their filings at eight to ten business days, 16.5 percent inside three, and a tail of 7.1 percent beyond fifteen. And the delay is counted in business days throughout; counting calendar days against a business-day rule, as Polk et al. (2024) do, misclassifies the 48 percent of their sample that piles at their day-ten marker.

6.3 The denominator and the audit

The result could be manufactured by measurement rather than by the rule, and the audit exists to close that off. Structured XML became mandatory on 2024-12-18, inside the post window, so post filings parse more easily than pre filings by construction; differential coverage alone would manufacture the primary result if the pre period simply failed to resolve more often. Gate G2 caps exactly this differential: the unresolved share is 2.4 percent pre against 0.7 percent post, a gap of 1.7 percentage points against a registered threshold of ten, and it passes. Gate G3 validates the parser itself: sixty filings, twenty from each non-empty period-by-route stratum (pre-text, post-text, post-xml), hand-coded before the parser's answer for those cases was seen, with the blind enforced by the order of the commits. The coded and parsed trigger dates disagree on zero of the sixty, against a registered threshold of three. The twenty xml-route excerpts in the audit sample read LABEL NOT FOUND, because the excerpt window targets a cover-page label that XML submissions do not carry; those cases were coded from the `dateOfEvent` element in the cached source document, the text a human coder would read.

Three data-quality items are stated rather than absorbed. Fourteen rows remain unresolved — 2.4 percent of the eligible pre population, 0.7 percent post — all on the text route with no recoverable trigger date. Three are Sageworth Trust Co. filings that name the filer's own CIK as the subject, form-type noise rather than campaigns; the other eleven are filings whose trigger date could not be recovered. Every estimate above carries all fourteen in its bounds. Ten of the 521 post filings declare a non-zero amendment number in the document body although the index lists them as originals; the population stays as registered — it is defined by the index form type — and these rows can only depress the post share. And there is no outcome screen: catch-up filings with extreme delays stay in the population, and the right tail they form is visible in Figure 2.

6.4 The headline

The headline is the filing-delay result of Table 2: after the February 2024 acceleration, the realised filing-delay distribution compressed sharply toward the five-business-day deadline, and the compression survives worst-case treatment of every unresolved filing. The statement is descriptive. It establishes that the statutory clock moved in practice; it says nothing about returns, activism, bidder entry, or takeover premia.

7 Conclusion

The disclosure rule is the market's partition, and this paper takes the identity literally. A stake threshold and a filing window split the histories that reach a control decision into a flagged cell and a pooled one; the window is a primitive of the model rather than a rescaling of noise, and the stake at filing is an object the model produces. On that ground the paper proves what it can and records where it cannot. Existence is a conditional whose two main hypotheses about the equilibrium object are measured to fail at the implemented calibration, stated once, in plain words, because a reader deserves to know exactly what the proposition does and does not cover. The attenuation theorem separates the rule's two margins honestly: the threshold margin attenuates with no dominance condition; the window margin attenuates if and only if a weight ratio times an unsigned composition ratio is at most one. The general-equilibrium survival of the fixed-policy sign is an implication with the region as a named hypothesis, plus numerical nodes. The numerical layer reads the same stack against the implemented calibration and reports an attenuation-direction window record alongside the measured failure of the chord restriction that would close the loop; the two facts sit in the same section because they are both true. The scope is stated as plainly as the results: the paper does not claim a global window-margin attenuation sign, κ-invariance of the filing-day jump, equilibrium uniqueness, a nonempty general-equilibrium region as a theorem, endogenous filing before the deadline, noisy or partially revealing flagged-round trading, continuous-time execution, or welfare or optimal rule design; nor that the predecessor model's non-monotone result for minority gains survives in the two-round model, nor that the predecessor calibration at $\Omega \approx 0.037$ is economically meaningful.

The empirics rest on the February 2024 acceleration. The established fact is descriptive: the statutory clock moved in practice. In the near-census of initial Schedule 13D filings, the median trigger-to-filing delay fell from six business days to five and the share filed within the new five-business-day deadline rose from 38.9 to 67.1 percent, a difference of 28.2 percentage points with a 95 percent bootstrap interval of 21.8 to 34.8; the difference survives assigning all fourteen unresolved filings to the extremes, and the registered coverage and parser gates passed. No control group supports the comparison, and it identifies no effect on liquidity, returns, or control outcomes.

Three open questions carry the research forward: whether any two-round pooled cell satisfies the chord restriction's support condition, which is the largest single conditionality in the stack; whether the existence assumptions can be satisfied on a relevant menu, including the incentive-compatibility question for a fully separating plan; and whether the scoped continuity problem in the outer-map assumption can be repaired constructively. Alongside them stands one open calibration input that is nowhere load-bearing: the disclosed share of engagements ω_a, for which no dataset in the literature offers an anchor, and which the paper does not invent. Each question is about the domain of an assumption this paper states rather than hides, which is where a theory of a legal rule should leave its unfinished work.

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